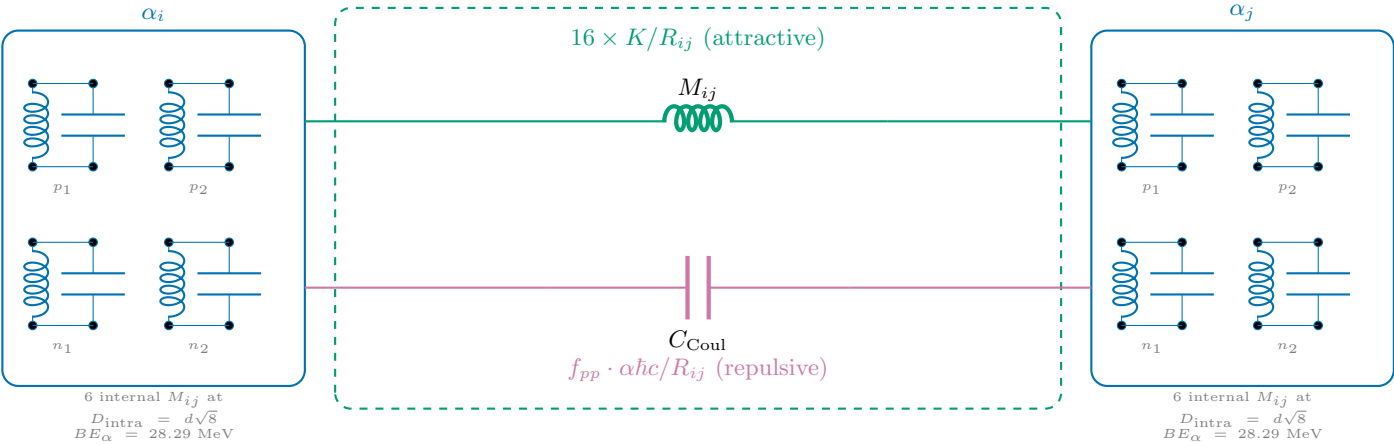


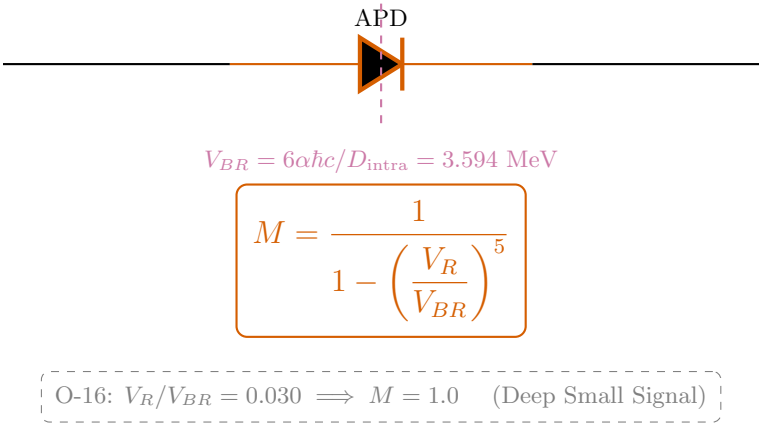
O-16 Semiconductor Equivalent Circuit

4α Tetrahedron — $V_R/V_{BR} = 0.030$ — $M = 1.000$ (Small Signal)

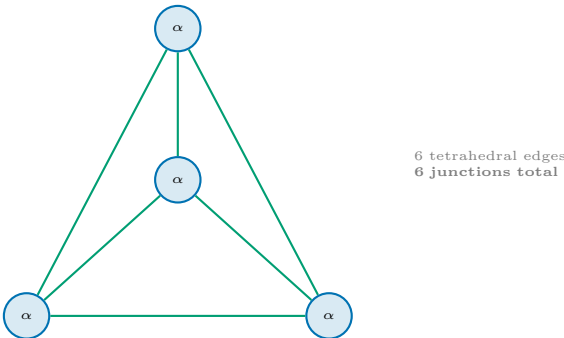
A: Inter-Alpha Junction (Unit Cell)



B: Miller Avalanche Stage



C: O-16 Topology (6 junctions)



D: Energy Balance

Energy balance equation:

$$M_{\text{nuc}} = 4 M_\alpha - BE_{\text{net}}$$

Net energy balance components:

$$BE_{\text{net}} = \underbrace{\sum_{i < j}^6 16 \cdot \frac{K}{R_{ij}}}_{\text{testStrong(attractive)}} - \underbrace{M \cdot \sum f_{pp} \frac{\alpha\hbar c}{R_{ij}}}_{\text{Coulomb (repulsive)}}$$

Strong: $\sum K/R$ = engine-derived

Coulomb: $\sum \alpha\hbar c/R \times f_{pp}$

Miller: $M = 1.000$ ($V_R/V_{BR} = 0.030$)

Result: 14,895.080 MeV (0.0000% error)

Each $\alpha = 2p + 2n$ LC tanks at tetrahedron vertices ($D_{\text{intra}} = d\sqrt{8} \approx 2.38$ fm).
 $d = 4\hbar/m_p c \approx 0.841$ fm. $K_{\text{mutual}} = 11.34$ MeV·fm. $V_{BR} = 3.594$ MeV.